

Homework 7

Markov random fields and energy-based models

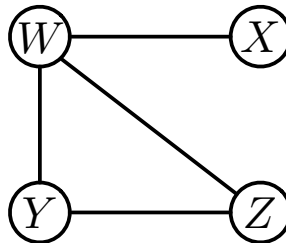
1. Probability distribution P over binary variables $x_1, \dots, x_6 \in \{0, 1\}$ has the following form:

$$P(x_1, \dots, x_6) = \frac{1}{Z} \psi(x_1, x_2) \psi(x_2, x_3) \psi(x_3, x_4) \psi(x_4, x_5) \psi(x_5, x_6) \psi(x_6, x_1)$$

where for $a, b \in \{0, 1\}$,

$$\psi(a, b) = \begin{cases} 1 & \text{if } a = b \\ 0.5 & \text{otherwise} \end{cases}$$

- (a) Draw the minimal undirected graph G over which P factors. (This is the undirected graph with the fewest possible edges that contains cliques corresponding to each of the terms in P .)
 - (b) For what configuration(s) $x \in \{0, 1\}^6$ is $P(x)$ maximized?
 - (c) For what configuration(s) $x \in \{0, 1\}^6$ is $P(x)$ minimized?
2. In a tournament, each player plays three games. Define three random variables:
- Random variable A is 1 if Tom wins his first game, and is 0 otherwise.
 - Random variable B is 1 if Tom wins his first two games, and is 0 otherwise.
 - Random variable C is 1 if Tom wins all three of his games, and is 0 otherwise.
- (a) Is C independent of A (i.e., is it true that for any values a, c that A, C might take on, we have $\Pr(A = a, C = c) = \Pr(A = a)\Pr(C = c)$)? Give a brief (one-sentence) reason for your answer.
Hint: The answer is no; show this by exhibiting specific choices of a, c for which $\Pr(A = a, C = c) \neq \Pr(A = a)\Pr(C = c)$.
 - (b) Is C conditionally independent of A given B (i.e., is it true that $\Pr(C = c|A = a, B = b) = \Pr(C = c|B = b)$)? Give a brief reason.
 - (c) Draw an undirected graphical model for (A, B, C) .
3. Four random variables W, X, Y, Z factor over the graph shown below.



- (a) What is the functional form of the joint distribution $P(W, X, Y, Z)$?
 (b) Write down a conditional independence property of this distribution.
4. The joint distribution P over (x_1, x_2, x_3) is given by the following table.

x_1	x_2	x_3	Pr
0	0	0	1/3
0	0	1	1/3
1	0	0	1/6
1	1	1	1/6

- (a) One of these variables is a *deterministic* function of the other two. Identify this relationship.
 (b) Draw the minimal Bayes net G over which this distribution factors. *Hint:* Start from (a), which shows that one variable is fully determined by the other two; are the latter two variables independent or not?
 (c) Now show how P factors over G by explicitly giving the three (conditional) probability tables, one per x_i .
 (d) Draw the minimal *undirected* graph G' over which distribution P factors.
5. *A Bayesian approach to image denoising.* Download the file `gibbs.zip` from the course website. This contains a Jupyter notebook and two images (`noisy_bear.png` and `original_bear.png`). The former is a corrupted version of the latter (which is provided only for reference).
 Let X be the corrupted image and Y the uncorrupted version that we wish to recover. Both are of the form $\{-1, +1\}^{M \times N}$. In the notebook, a Markov random field joint distribution for (X, Y) is given. You will use this to design a Gibbs sampler for $Y|X$.
- (a) Let p denote a specific pixel in the image. Then Y_p is the (reconstructed) value at that pixel, and $Y_{\setminus p}$ is the value at all remaining pixels. Give a precise expression for $\Pr(Y_p = +1 | Y_{\setminus p}, X)$.
 (b) Implement a Gibbs sampler at the appropriate place in the notebook. Initialize Y by setting each pixel to $\{-1, +1\}$ uniformly at random. Then apply the sampler and show the image Y at various stages while the Markov chain is mixing: show any **five** of the images obtained along the way.
 (c) To reconstruct the image, just set the value of each pixel p to $\mathbb{E}[Y_p | X]$. Show this image.
6. *Rejection sampling.* Suppose we want to sample from a one-dimensional density $f(x)$ for which we do not have a readymade sampler. One way to do this is via *rejection sampling*. Here's the idea:

- We choose another distribution $g(x)$ from which we know how to sample (e.g. uniform or Gaussian distribution). We will call this the *proposal distribution*. It is important that $g(x)$ be nonzero whenever $f(x)$ is nonzero.
- Let M be an upper bound on $f(x)/g(x)$; that is, $f(x)/g(x) \leq M$ for all x .
- To generate a sample from f , we repeat the following procedure until it *accepts*:
 - Generate a sample x from g
 - With probability $f(x)/(Mg(x))$, *accept* the sample. Otherwise, *reject* it.

Let's see how this works in practice. Let's say we want to sample a number in the range $[0, \pi]$ from

$$f(x) = \frac{1}{2} \sin x$$

(think of x as being in radians).

- (a) Plot the density of f .
- (b) Give a suitable proposal distribution $g(x)$. What is a suitable upper bound M on $f(x)/g(x)$?
- (c) Use rejection sampling to generate 10000 samples from f . How many times did you need to sample from g ?
- (d) Plot the histogram of the samples you generated, using 50 bins. It should bear a resemblance to the density in (a).